

DFS Management and File Server Resource Manager (FSRM)

Implementation Documentation

1. Introduction

This document describes the implementation and configuration of **Distributed File System (DFS) Management** and **File Server Resource Manager (FSRM)** in the **mrt.com** domain. The goal of this setup is to provide **centralized access to shared folders**, ensure **data availability through replication**, and enforce **storage and file-type controls**.

2. Environment Overview

Component	Details
Domain Name	mrt.com
Domain Controller	ad.mrt.com
Primary File Server	fs01.mrt.com
Secondary / Replica Server	ad.mrt.com
Server OS	Windows Server 2019

3. DFS Management Overview

Purpose of DFS

DFS was implemented to:

- Provide a **single logical namespace** for shared folders
- Hide physical server locations from users
- Ensure **data availability using replication**
- Improve file service reliability

4. DFS Namespace Configuration

Item	Details
Namespace Type	Domain-based DFS Namespace
Namespace Domain	mrt.com

Item	Details
Namespace Host	ad.mrt.com

Users access shared data through a single DFS path instead of accessing servers directly.

5. DFS Shared Folders and Replication Setup

5.1 DFS Folder Targets

DFS Folder Name	Server	Physical Share
file	ad01.mrt.com	\\ad\file
files1	fs01.mrt.com	\\fs01\files1

5.2 DFS Replication Configuration

Item	Configuration
Primary (Active) Server	fs01.mrt.com
Replication Partner	ad.mrt.com
Replication Type	DFS Replication (DFSR)
Initial Data Source	FS01

Replication Behavior:

- **FS01** acts as the **primary server**.
- Data changes originate from **FS01**.
- Changes are replicated to **AD**.
- In case FS01 becomes unavailable, data remains accessible from AD01.

6. File Server Resource Manager (FSRM) Configuration

6.1 Purpose of FSRM

FSRM was implemented to:

- Control the types of files stored on servers
- Enforce storage limits
- Prevent misuse of shared storage

7. FSRM Shared Folder Configuration

Item	Details
Shared Folder Name	files2
Server	fs01.mrt.com
Purpose	Controlled storage with quota and file screening

8. File Screening Policy (Blocking Executable Files)

Configuration Details:

- **Active File Screening** enabled on **files2**
- Executable files were blocked to improve security

Setting	Value
Blocked File Type	.exe
Screening Mode	Active
Applied Folder	files2

Result:

- Users cannot copy or create .exe files
- Restricted files are blocked immediately

9. Quota Management Configuration

Quota Details:

Item	Value
Quota Type	Hard Quota
Quota Limit	2 GB
Drive Letter	Z:
Assigned User	rms1

Notes:

- User **rms1** is restricted to **2 GB** storage.
- Storage usage beyond the quota is blocked.
- Quota enforcement is handled by FSRM.

10. Access and Control Summary

User	Resource	Access
rms1	DFS Shares	Allowed
rms1	files2 (Z:)	2 GB Hard Quota
rms1	Executable Files (.exe)	Blocked
rms2	DFS Shares	Allowed (based on permissions)

11. Testing and Validation

Tests Performed:

1. Accessed DFS namespace from client machine.
2. Verified data availability from **FS01** and **AD01**.
3. Copied test data to confirm DFS replication.
4. Attempted to upload .exe files to **files2**.
5. Filled storage to test quota enforcement.

Results:

- DFS replication worked correctly from **FS01** → **AD01**
- .exe files were blocked successfully
- Quota limit stopped storage usage after **2 GB**
- No replication or access errors observed

12. Security and Operational Benefits

- High availability through DFS replication
- Centralized and simplified file access
- Protection against unauthorized executables
- Controlled storage usage per user
- Improved data governance

13. Conclusion

The **DFS Management** and **File Server Resource Manager (FSRM)** implementation in the **mrt.com** domain was completed successfully.

The **primary file server (FS01)** replicates data to **AD01**, ensuring availability and resilience. FSRM policies effectively enforce **file-type restrictions** and **storage quotas**, improving overall security and storage management.